VIVACOM

ДЗ задачи:

① $\int_0^{1/2} \arcsin x\,dx = x\cdot\arcsin x\Big|_0^{\frac{1}{2}} - \int x\,d\arcsin x =$

$= \frac{1}{2}\cdot\arcsin\frac{1}{2} - \int_0^{\frac{1}{2}} \frac{x}{\sqrt{1-x^2}}\,dx =$

$= \frac{1}{2}\cdot\frac{\pi}{6} + \frac{1}{2}\int_0^{1/2} \frac{d-x^2}{\sqrt{1-x^2}} = \frac{\pi}{12} + \frac{1}{2}\int_0^{1/2} \frac{d(1-x^2)}{\sqrt{1-x^2}} =$

$= \frac{\pi}{12} + \frac{1}{2}\int_0^{1/2} (1-x^2)^{-\frac{1}{2}}\,d(1-x^2) = \frac{\pi}{12} + \frac{1}{2}\frac{(1-x^2)^{-\frac{1}{2}+1}}{-\frac{1}{2}+1}\Big|_0^{\frac{1}{2}}$

$\frac{\pi}{12} + \frac{1}{2}\cdot 2(1-x^2)^{\frac{1}{2}}\Big|_0^{\frac{1}{2}} = \frac{\pi}{12} - 2\left(\sqrt{1-\frac{1}{4}} - \sqrt{1-0}\right) =$

$= \frac{\pi}{12} + 1,5\left(\frac{\sqrt{3}}{2} - 1\right) = \frac{\pi}{12} + \frac{\sqrt{3}}{2} - 1$

Юни 2015г. ▷

② Юни 2016г.

$\int_1^2 x\log_2\frac{1}{x}\,dx = \int_1^2 x\cdot\frac{\ln\frac{1}{x}}{\ln 2}\,dx =$ $\left\|\begin{array}{l}\ln x^{-1} = \\ -\ln x \\ \ln x^2 = 2\ln x \\ \log_2 2^3 = 3 \\ \log_2 a^3 = 3\log_2 a\end{array}\right.$

$= \frac{1}{\ln 2}\int_1^2 x\cdot\ln\frac{1}{x} = \frac{1}{\ln 2}\int_1^2 x\cdot\ln x^{-1}\,dx =$

$= -\frac{1}{\ln 2}\int_1^2 x\cdot\ln x\,dx =$

$= -\frac{1}{\ln 2}\cdot\frac{1}{2}\int_1^2 \ln x\cdot dx^2 = -\frac{1}{\ln 2}\cdot\frac{1}{2}\left[x^2\cdot\ln x\Big|_1^2 - \int_1^2 x^2\,d\ln x\right] =$

$= -\frac{1}{2\ln 2}\left[4\cdot\ln 2 - 1\cdot\ln 1 - \int_1^2 x^2\cdot\frac{1}{x}\,dx\right] =$

$= -\frac{1}{2\ln 2}\cdot\left[4\ln 2 - 0 - \int_1^2 x\,dx\right] = -\frac{1}{2\ln 2}\left[4\ln 2 - \frac{x^2}{2}\Big|_1^2\right] =$

$= -2 + \frac{1}{2\ln 2}\cdot\frac{1}{2}(4-1) = -2 + \frac{3}{4}\cdot\frac{1}{\ln 2}$ ▷

$$\int_0^1 x^2 \operatorname{arctg} x\,dx = \frac{1}{3}\int_0^1 \operatorname{arctg} x\,dx^3 =$$

$$= \frac{1}{3}\cdot\left[x^3 \operatorname{arctg} x\Big|_0^1 - \int_0^1 x^3\, d\operatorname{arctg} x\right] = \frac{1}{3}\cdot\left[1\cdot\operatorname{arctg} 1 - 0 - \int_0^1 \frac{x^3}{1+x^2}\,dx\right]$$

$$= \frac{1}{3}\cdot\left[1\cdot\frac{\pi}{4} - \underbrace{\int_0^1 \frac{x^3+x-x}{1+x^2}\,dx}_{I}\right]$$

$\frac{\pi}{4}$

$$I = \int_0^1 \frac{x^3+x}{1+x^2}dx - \int_0^1 \frac{x}{1+x^2}dx =$$

$$= \int_0^1 \frac{x(x^2+1)}{x^2+1}dx - \frac{1}{2}\int_0^1 \frac{d(x^2+1)}{1+x^2}dx =$$

$$= \frac{x^2}{2}\Big|_0^1 - \frac{1}{2}\ln(1+x^2)\Big|_0^1 =$$

$$= \frac{1}{2} - \frac{1}{2}\left(\ln 2 - \ln 1\right) = \frac{1}{2} - \frac{\ln 2}{2}$$

$$\Rightarrow \frac{1}{3}\cdot\left[\frac{\pi}{4} - \frac{1}{2} + \frac{\ln 2}{2}\right] = \frac{\pi}{12} - \frac{1}{6} + \frac{\ln 2}{6} \quad \square$$

③ Септември 2019г.

$$\int_0^2 \ln(x^2+4)\,dx = x\cdot\ln(x^2+4)\Big|_0^2 - \int_0^2 x\, d\ln(x^2+4) =$$

$$= 2\cdot\ln 8 - 0 - \int_0^2 \frac{x\cdot 2x}{x^2+4}dx = 2\ln 8 - 2\int_0^2 \frac{x^2}{x^2+4}dx$$

$$= 2\ln 8 - 2\int_0^2 \frac{x^2+4-4}{x^2+4}dx =$$

$$= 2\ln 8 - 2\int_0^2 \frac{x^2+4}{x^2+4} + 8\int_0^2 \frac{1}{x^2+4}dx =$$

$$= 2\ln 8 - 2x\Big|_0^2 + 8\int_0^2 \frac{1}{x^2+2^2}dx =$$

$$= 2\ln 8 - 2\cdot(2-0) + 8\cdot\frac{1}{2}\operatorname{arctg}\frac{x}{2}\Big|_0^2 = 2\ln 8 - 4 + 4(\operatorname{arctg} 1 - \operatorname{arctg} 0)$$

$$= 2\ln 8 - 4 + 4\left(\frac{\pi}{4} - 0\right) = 2\ln 8 - 4 + \pi$$

$\square$

$\operatorname{tg} x = \frac{\sin x}{\cos x}$

$\sin^2 x$

VIVACOM Септември 2018

$$I = \int_0^{\pi/2} x\cos^2 x\,dx = \int_0^{\pi/2} x\cdot\left(\frac{1+\cos 2x}{2}\right)dx =$$

$$= \frac{1}{2}\int_0^{\pi/2} x\,dx + \frac{1}{2}\underbrace{\int_0^{\pi/2} x\cdot\cos 2x\,dx}_{J}$$

$$J = \int_0^{\pi/2} x\cdot\cos 2x\,dx = \frac{1}{2}\int_0^{\pi/2} x\,d\sin 2x =$$

$$= \frac{1}{2}\left[x\cdot\sin 2x\Big|_0^{\frac{\pi}{2}} - \int_0^{\pi/2}\sin 2x\,dx\right] =$$

$$= \frac{1}{2}\left[\frac{\pi}{2}\cdot\sin\pi - 0 + \frac{1}{2}\cos 2x\Big|_0^{\frac{\pi}{2}}\right] =$$

$$= \frac{1}{2}\cdot\left[\frac{\pi}{2}\cdot 0 - 0 + \frac{1}{2}\cdot(\cos\pi - \cos 0)\right] =$$

$$= \frac{1}{2}\cdot\frac{1}{2}\cdot(-1-1) = \frac{1}{4}\cdot(-2) = -\frac{1}{2}$$

$$\Rightarrow I = \frac{1}{2}\cdot\frac{x^2}{2}\Big|_0^{\frac{\pi}{2}} + \frac{1}{2}\cdot\left(-\frac{1}{2}\right) = \frac{1}{4}\cdot\frac{\pi^2}{4} - \frac{1}{4} = \frac{\pi^2}{16} - \frac{1}{4}$$ □

зад:

$$I = \int e^x\sin x\,dx = \int \sin x\,d e^x =$$

$$= e^x\cdot\sin x - \int e^x\,d\sin x = e^x\cdot\sin x - \int e^x\cdot\cos x\,dx + C$$

$$= e^x\cdot\sin x - \int\cos x\,d e^x = e^x\cdot\sin x - e^x\cdot\cos x + \int e^x\,d\cos x + C$$

$$= e^x\cdot\sin x - e^x\cdot\cos x - \underbrace{\int e^x\cdot\sin x\,dx}_{I} + C$$

$$I = e^x\cdot\sin x - e^x\cdot\cos x - I + C$$

$$2I = e^x\cdot\sin x - e^x\cdot\cos x + C$$

$$I = \frac{e^x\cdot\sin x - e^x\cdot\cos x + C}{2}$$ □

$\int x \cdot \sin x \cdot \cos 2x \, dx$ // $\sin\alpha \cdot \cos\beta = \frac{\sin(\alpha+\beta)+\sin(\alpha-\beta)}{2}$ // $\sin\alpha = -\sin x$

$\int x \cdot \left(\frac{\sin(x+2x)+\sin(x-2x)}{2}\right) dx = \frac{1}{2}\int x \cdot \sin 3x \, dx + \frac{1}{2}\int x(-\sin x)\, dx$

$= -\frac{1}{2}\cdot\frac{1}{3}\int x \cdot d\cos 3x + \frac{1}{2}\int x \cdot d\cos x =$

$= -\frac{1}{6}\left[x\cdot\cos 3x - \int \cos 3x\, dx\right] + \frac{1}{2}\left[x\cdot\cos x - \int \cos x\, dx\right] + C$

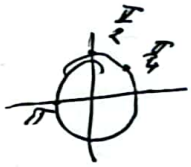
$= -\frac{1}{6}x\cdot\cos 3x + \frac{1}{6}\cdot\frac{1}{3}\cdot\sin 3x + \frac{1}{2}x\cdot\cos x - \frac{1}{2}\sin x + C$

$= \frac{1}{18}\cdot\left(-9\sin x + \sin 3x + 9x\cdot\cos x - 3x\cdot\cos 3x\right) + C$ □

$\int_0^{\frac{\pi}{2}} \left|\frac{1}{2} - \sin^2 x\right| dx$

$\frac{1}{2} - \left(\frac{1-\cos 2x}{2}\right) = \frac{1}{2} - \frac{1}{2} + \frac{\cos 2x}{2}$

$\int_0^{\frac{\pi}{2}} \left|\frac{1}{2} - \sin^2 x\right| dx = \int_0^{\frac{\pi}{2}} \left|\frac{\cos 2x}{2}\right| dx =$



$= \int_0^{\frac{\pi}{4}} \left|\frac{\cos 2x}{2}\right| dx + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \left|\frac{\cos 2x}{2}\right| dx =$

$= \int_0^{\frac{\pi}{4}} \frac{\cos 2x}{2}\, dx + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} -\frac{\cos 2x}{2}\, dx =$

$= \frac{1}{2}\cdot\frac{1}{2}\sin 2x\Big|_0^{\frac{\pi}{4}} + \frac{1}{2}\cdot\frac{1}{2}\cdot\sin 2x\Big|_{\frac{\pi}{4}}^{\frac{\pi}{2}} =$

$= \frac{1}{4}\cdot\left(\sin\frac{\pi}{2} - \sin 0\right) - \frac{1}{4}\left(\sin\pi - \sin\frac{\pi}{2}\right)$

$= \frac{1}{4}\cdot(1-0) - \frac{1}{4}(0-1) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$ □

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Септември 2015г.

$$\int \frac{x-2}{x(x^2+2)}\,dx = \int \frac{x-2}{x^3+2x}\,dx$$

$$\frac{x-2}{x(x^2+2)} = \frac{A}{x} + \frac{Bx+C}{x^2+2} = \frac{Ax^2+2A+Bx^2+Cx}{x(x^2+2)} =$$

$$= \frac{(A+B)x^2+Cx+2A}{x(x^2+2)} =$$

$$\Rightarrow \begin{cases} A+B=0 \\ C=1 \\ 2A=-2 \end{cases} \Rightarrow A=-1 \Rightarrow B=1$$

$$\Rightarrow \frac{x-2}{x(x^2+2)} = -\frac{1}{x} + \frac{x+1}{x^2+2}$$

Проверка: $-\frac{1}{x} + \frac{x+1}{x^2+2} = \frac{-x^2-2+x^2+x}{x(x^2+2)} = \frac{x-2}{x(x^2+2)}$ ✓

$$\int -\frac{1}{x} + \frac{x+1}{x^2+2}\,dx = -\ln x + \int \frac{x}{x^2+2}\,dx + \int \frac{1}{x^2+2}\,dx + C$$

$$= -\ln|x| + \frac{1}{2}\int \frac{d(x^2+2)}{x^2+2} + \frac{1}{\sqrt{2}}\operatorname{arctg}\frac{x}{\sqrt{2}}\,dx + C =$$

$$= -\ln|x| + \frac{1}{2}\ln(x^2+2) + \frac{\sqrt{2}}{2}\operatorname{arctg}\frac{\sqrt{2}x}{2} + C \quad \square$$

метод на
-Неопределени коефициенти - пример

$$\frac{R(x)}{(x-2)\cdot(x+3)^3(x^2+2x+4)^2}$$

$$\frac{A}{x-2} + \frac{B}{(x+3)^3} + \frac{Cx+D}{(x+3)^2} + \frac{Ex^2+Fx+G}{}$$

$$\frac{A}{x-2} + \frac{B}{x+3} + \frac{C}{(x+3)^2} + \frac{E}{(x+3)^3} + \frac{Fx+G}{x^2+2x+4} + \frac{Hx+I}{x^2+2x+4}$$

НОли 2019г.

$$\int_1^8 \frac{dx}{\sqrt[3]{x}+x} \quad \frac{t=\sqrt[3]{x}}{x=t^3} \quad \int_1^2 \frac{dt^3}{t+t^3} = \int_1^2 \frac{3t^2\,dt}{t+t^3} =$$

$$= 3\int_1^2 \frac{t^2\,dt}{t(1+t^2)} = 3\cdot\frac{1}{2}\int_1^2 \frac{d(1+t^2)}{1+t^2} = \frac{3}{2}\cdot \ln(1+t^2)\Big|_1^2 =$$

$$= \frac{3}{2}\cdot(\ln(5)-\ln 2)$$ □

$$\int_0^{\pi/2} \frac{1}{2+\sin x}\,dx$$

Прилагаме универсална субституция:

$x = 2\operatorname{arctg} t$; $t = \operatorname{tg}\frac{x}{2}$; $\sin x = \frac{2t}{1+t^2}$

$dx = \frac{2}{1+t^2}$; $2\operatorname{arctg}(\alpha) = 0 \Rightarrow \alpha = 0$

$2\operatorname{arctg}(\beta) = \frac{\pi}{2} \Rightarrow \beta =$

$\operatorname{arctg}(\beta) = \frac{\pi}{4} \Rightarrow \beta = 1$



У-с

$$\Rightarrow \int_0^1 \frac{\frac{2}{1+t^2}}{\left(2+\frac{2t}{1+t^2}\right)\cdot(1+t^2)}\,dt =$$

$$= \int_0^1 \frac{2}{2+2t+2t^3}\,dt = \int_0^1 \frac{2(1+t^2)}{2+2t+2t^3}\,dt$$

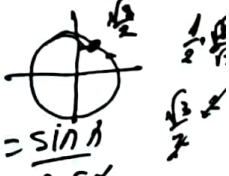
$$\int_0^1 \frac{2}{2+2t^2+2t}\,dt = \int_0^1 \frac{1}{1+t+t^2}\,dt =$$

$$= \int_0^1 \frac{1}{\left(t+\frac{1}{2}\right)^2+\frac{3}{4}}\,dt = \int_0^1 \frac{d\left(t+\frac{1}{2}\right)}{\left(t+\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2} =$$

$$= \frac{1}{\frac{\sqrt{3}}{2}}\operatorname{arctg}\left(\frac{t+\frac{1}{2}}{\frac{\sqrt{3}}{2}}\right)\Big|_0^1 = \frac{2}{\sqrt{3}}\cdot\left(\operatorname{arctg}\left(\frac{1+\frac{1}{2}}{\frac{\sqrt{3}}{2}}\right)-\operatorname{arctg}\left(\frac{\frac{1}{2}}{\frac{\sqrt{3}}{2}}\right)\right)$$

$$= \frac{2}{\sqrt{3}}\cdot\left(\operatorname{arctg}\left(\frac{3}{2}\cdot\frac{2}{\sqrt{3}}\right)-\operatorname{arctg}\left(\frac{1}{2}\cdot\frac{2}{\sqrt{3}}\right)\right) = \frac{2}{\sqrt{3}}\cdot\left(\operatorname{arctg}\sqrt{3}-\operatorname{arctg}\frac{\sqrt{3}}{3}\right)$$

$$= \frac{2}{\sqrt{3}}\cdot\left(\frac{\pi}{3}-\frac{\pi}{6}\right)$$ □



$\operatorname{tg} x = \frac{\sin x}{\cos x}$

$\operatorname{tg} x = \sqrt{3}$

$(x^2+2x+1)' = 2x+2$

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$$\int \frac{3x+2}{x^2+2x+4}\,dx = \int \frac{3x+2+1-1}{(x+1)^2+3}\,dx =$$

$$= \int \frac{3x+3}{(x+1)^2+3}\,dx - \int \frac{1}{(x+1)^2+3}\,d(x+1) =$$

$$= \frac{3}{2}\int \frac{d\left((x+1)^2+3\right)}{(x+1)^2+3} - \frac{1}{\sqrt{3}}\operatorname{arctg}\left(\frac{x+1}{\sqrt{3}}\right) =$$

$$= \frac{3}{2}\ln\left((x+1)^2+3\right) - \frac{1}{\sqrt{3}}\operatorname{arctg}\left(\frac{x+1}{\sqrt{3}}\right)$$

08.09.2021г.

$$\int \operatorname{arctg}\sqrt{x}\,dx \quad \begin{matrix} y=\sqrt{x} \\ x=y^2 \end{matrix} \quad \int \operatorname{arctg} y\,dy^2 =$$

$$y^2\cdot\operatorname{arctg} y - \int y^2 d\operatorname{arctg} y + C =$$

$$= y^2\cdot\operatorname{arctg} y - \int \frac{y^2+1-1}{1+y^2}\,dy + C =$$

$$= y^2\cdot\operatorname{arctg} y - \int \frac{y^2+1}{1+y^2}\,dy + \int \frac{1}{1+y^2}\,dy + C =$$

$$= y^2\cdot\operatorname{arctg} y - y + \operatorname{arctg} y + C$$

$$= x\cdot\operatorname{arctg}\sqrt{x} - \sqrt{x} + \operatorname{arctg}\sqrt{x} + C$$

$$= (x+1)\operatorname{arctg}\sqrt{x} - \sqrt{x} + C \qquad \square$$

13.07.2021г.

$$I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} e^{|x|}(\sin x + \cos x)\,dx$$

$$e^{|x|} = \begin{cases} e^x, & x \in \left[-\frac{\pi}{2}, 0\right] \\ e^{-x}, & x \in \left(0, \frac{\pi}{2}\right] \end{cases}$$

$$\Rightarrow I = \underbrace{\int_{-\frac{\pi}{2}}^{0} e^{-x}(\sin x + \cos x)\,dx}_{I} + \underbrace{\int_{0}^{\frac{\pi}{2}} e^{x}(\sin x + \cos x)}_{Q} =$$

$$I = \int_{-\frac{\pi}{2}}^{0} e^{-x}(\sin x + \cos x)\,dx = \underbrace{\int_{-\frac{\pi}{2}}^{0} e^{-x}\cdot \sin x\,dx}_{J_2} + \int_{-\frac{\pi}{2}}^{0} e^{-x}\cdot\cos x\,dx$$

$$J_1 = \int_{-\frac{\pi}{2}}^{0} e^{-x}\cdot\sin x\,dx = -\int_{-\frac{\pi}{2}}^{0} \sin x\,d e^{-x}$$

$$= -e^{-x}\cdot\sin x\Big|_{-\frac{\pi}{2}}^{0} + \int_{-\frac{\pi}{2}}^{0} e^{-x}\cdot d\sin x = -\left(e^{0}\cdot\sin 0 - e^{+\frac{\pi}{2}}\cdot\sin\left(-\frac{\pi}{2}\right)\right) + \int_{-\frac{\pi}{2}}^{0} e^{-x}\cdot\cos x\,dx$$

$$= -\left(1\cdot 0 - e^{\frac{\pi}{2}}\cdot(-1)\right) + \int_{-\frac{\pi}{2}}^{0} \cos x\cdot d\cos x =$$

$$= -e^{\frac{\pi}{2}} - \left(\cos x\cdot e^{-x}\Big|_{-\frac{\pi}{2}}^{0}\right) + \int_{-\frac{\pi}{2}}^{0} e^{-x}\cdot d\cos x =$$

$$= -e^{\frac{\pi}{2}} - \left(\cos 0\cdot e^{0} - \cos\left(-\frac{\pi}{2}\right)\cdot e^{\frac{\pi}{2}}\right) - \int_{-\frac{\pi}{2}}^{0} e^{-x}\cdot\sin x\,dx$$

$$= J_1$$

$$2J_1 = -e^{\frac{\pi}{2}} - (1\cdot 1 - 0\cdot e^{-\frac{\pi}{2}})$$

$$J_1 = \frac{-e^{\frac{\pi}{2}} - 1}{2}$$

Аналогично за останалите.

По-хитро решение:

$$I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} e^{|x|}(\sin x + \cos x)\,dx = \underbrace{\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} e^{|x|}\cdot\sin x\,dx}_{\text{нечетна}} + \underbrace{\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} e^{|x|}\cdot\cos x\,dx}_{\text{четна}}$$

но функцията $f(x) = e^{|x|}\cdot\sin x$ е нечетна $\Rightarrow (f(x) = -f(x)) \Rightarrow$

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} e^{|x|}\cdot\sin x\,dx = 0$$

функцията $g(x) = e^{|x|}\cdot\cos x$ е четна $g(x) = g(-x) \Rightarrow$

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} e^{|x|}\cdot\cos x\,dx = 2\int_{0}^{\frac{\pi}{2}} e^{|x|}\cdot\cos x\,dx$$

но в интервала $[0; \frac{\pi}{2}]$, $e^{|x|} = e^{x}$

$$\Rightarrow I = 2\int_{0}^{\frac{\pi}{2}} e^{x}\cdot\cos x\,dx$$

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$$I = 2\int_0^{\frac{\pi}{2}} \cos x \, d e^x = 2\cdot\left(\cos x \cdot e^x \Big|_0^{\frac{\pi}{2}} - \int_0^{\pi/2} e^x \, d\cos x\right) =$$

$$= 2\cdot\left(\underset{0}{\cos\frac{\pi}{2}} \cdot e^{\frac{\pi}{2}} - \underset{1}{\cos 0}\cdot \underset{1}{e^0} + \int_0^{\frac{\pi}{2}} \sin x \, d e^x\right) =$$

$$= 2\left(-1 + \sin x \cdot e^x \Big|_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} e^x \cdot d\sin x\right) =$$

$$= 2\cdot\left(-1 + \underset{1}{\sin\frac{\pi}{2}}\cdot e^{\frac{\pi}{2}} - \underset{0}{\sin 0}\cdot \underset{1}{e^0} - \underbrace{\int_0^{\frac{\pi}{2}} e^x \cdot \cos x \, dx}_{\frac{1}{2}I}\right) =$$

$$= -2 + 2e^{\frac{\pi}{2}} - I$$

$$2I = -2 + 2e^{\frac{\pi}{2}}$$

$$I = e^{\frac{\pi}{2}} - 1 \qquad \text{Отг: } e^{\frac{\pi}{2}} - 1 \square$$

12.07.2022г.

$$I = \int_0^1 \sqrt{x}\ln\sqrt{1+x}\, dx = \frac{2}{3}\int_0^1 \ln\sqrt{1+x}\, d x\sqrt{x}$$

$$= \frac{2}{3}\ln\sqrt{1+x}\cdot x\sqrt{x}\Big|_0^1 - \frac{2}{3}\int_0^1 x\sqrt{x}\, d\ln\sqrt{1+x} =$$

$$= \frac{2}{3}\left(\ln\sqrt{2}\cdot 1 - \ln 1\cdot 0\right) - \frac{2}{3}\int_0^1 \frac{x\cdot\sqrt{x}\cdot\left(\sqrt{1+x}\right)'}{\sqrt{1+x}}\, dx$$

$$= \frac{2}{3}\ln\sqrt{2} - \frac{2\cdot 1}{3\cdot 2}\int_0^1 \frac{x\sqrt{x}}{\sqrt{1+x}\cdot\sqrt{1+x}}\, dx =$$

$$= \frac{2}{3}\ln\sqrt{2} - \frac{1}{3}\underbrace{\int_0^1 \frac{x\sqrt{x}}{1+x}\, dx}_{J}$$

$$J = \int_0^1 \frac{x\sqrt{x}}{1+x}\, dx \overset{x=t^2}{=\!=} \int_0^1 \frac{t^2\cdot t}{1+t^2}\, d t^2 =$$

$$= 2\int_0^1 \frac{t^4+1-1}{1+t^2}\, dt = 2\int_0^1 \frac{(t^2-1)(t^2+1)}{1+t^2}\, dt + 2\int_0^1 \frac{1}{t^2+1}\, dt$$

$= 2\int_0^1 t^2\,dt - 2\int_0^1 dt + 2\operatorname{arctg} t\Big|_0^1 =$

$= 2\cdot\frac{t^3}{3}\Big|_0^1 - 2\cdot 1 + 2(\operatorname{arctg}1 - \operatorname{arctg}0) =$

$= \frac{2}{3} - 2 + 2\left(\frac{\pi}{4} - 0\right) = \frac{2}{3} - \frac{6}{3} + \frac{\pi}{2} = -\frac{4}{3} + \frac{\pi}{2}$

$\Rightarrow I = \frac{2}{3}\ln 2 - \frac{1}{3}\left(\frac{4}{3} + \frac{\pi}{2}\right) = \frac{2}{3}\ln 2 - \frac{4}{9} - \frac{\pi}{6}$

Вторы начин:

$I = \int_0^1 \sqrt{x}\ln\sqrt{x+1}\,dx =$

$= \frac{1}{2}\cdot\int_0^1 \sqrt{x}\ln(1+x)\,dx \overset{x=y^2}{=} \frac{1}{2}\int_0^1 y\ln(1+y^2)\,dy^2 =$

$= \frac{1}{2}\cdot 2\cdot\int_0^1 y\cdot y\ln(1+y^2)\,dy =$

$= \frac{1}{3}\int_0^1 \ln(1+y^2)\,dy^3 = \frac{1}{3}\cdot\left[\ln(1+y^2)\cdot y^3\Big|_0^1 - \int_0^1 y^3\,d\ln(1+y^2)\right]$

$= \frac{1}{3}\cdot\left[\ln 2\cdot 1 - 0 - \int_0^1 \frac{y^3\cdot 2y}{1+y^2}\,dy\right] =$

$= \frac{1}{3}\cdot\left[\ln 2 - 2\int_0^1 \frac{y^4+1-1}{1+y^2}\,dy\right] =$

$= \frac{1}{3}\cdot\left[\ln 2 - 2\int_0^1 \frac{(y^2-1)\cdot(y^2+1)}{1+y^2}\,dy + 2\int_0^1 \frac{1}{y^2+1}\,dy\right] =$

$= \frac{1}{3}\cdot\left[\ln 2 - 2\left(\frac{y^3}{3}\Big|_0^1 - y\Big|_0^1\right) - 2\operatorname{arctg} y\Big|_0^1\right] =$

$= \frac{1}{3}\left[\ln 2 - 2\left(\frac{1}{3} - 0 - 1 + 0\right) - 2\left(\operatorname{arctg}1 - \operatorname{arctg}0\right)\right] =$

$= \frac{1}{3}\left(\ln 2 + \frac{4}{3} - \frac{2\pi}{2}\right)$

D

VIVACOM

$I = \int (x + tg^2 x) \sin^2 x \, dx \quad , x \in \left(\frac{\pi}{2}, \frac{\pi}{2}\right)$

$= \int x \cdot \sin^2 x \, dx + \int \frac{\sin^2 x}{\cos^2 x} \cdot (1 - \cos^2 x) \, dx$

$= \int x \cdot \left(\frac{1 - \cos 2x}{2}\right) dx + \int \frac{\sin^2 x}{\cos^2 x} dx - \int \sin^2 x \, dx =$

$= \frac{1}{2} \int x - x \cdot \cos 2x \, dx + \int \sin^2 x \, d tg x - \int \frac{1 - \cos 2x}{2} dx =$

$= \frac{1}{2} \int x - \frac{1}{2} \int x \cos 2x \, dx +$

$tg^2 x \cdot \sin^2 x = \frac{\sin^2 x}{\cos^2 x} \cdot \sin^2 x = \frac{1}{\cos^2 x} \cdot (\sin^2 x \cdot \sin^2 x) =$

$= \frac{1}{\cos^2 x} \cdot (1 - \cos^2 x)^2 = \frac{1}{\cos^2 x}$

$tg^2 x \cdot \sin^2 x = tg^2 x \left(1 - \cos^2 x\right) = tg^2 x - tg^2 x \cdot \cos^2 x =$

$= tg^2 x - \sin^2 x =$

$\frac{\sin^2 x}{\cos^2 x} - \frac{1 - \cos^2 x}{\cos^2 x} =$

$tg^2 x \cdot \sin^2 x = tg^2 x \left(1 - \cos^2 x\right) = tg^2 x - \frac{\sin^2 x}{\cos^2 x} \cdot \cos^2 x =$

$= tg^2 x - \sin^2 x = \frac{\sin^2 x}{\cos^2 x} - \left(\frac{1 - \cos 2x}{2}\right) =$

$= \frac{1 - \cos^2 x}{\cos^2 x} - \frac{1}{2} + \frac{\cos 2x}{2} = \frac{1}{\cos^2 x} - 1 - \frac{1}{2} + \frac{\cos 2x}{2} =$

$= \frac{1}{\cos^2 x} - \frac{3}{2} + \frac{\cos 2x}{2}$

$\Rightarrow I = \int x \cdot \sin^2 x \, dx + \int tg^2 x \cdot \sin^2 x \, dx =$

$= \int x \cdot \left(\frac{1 - \cos 2x}{2}\right) dx + \int \frac{1}{\cos^2 x} - \frac{3}{2} + \frac{\cos 2x}{2} dx =$

$= \frac{1}{2}\int x\,dx - \frac{1}{2}\int x\cdot\cos 2x\,dx + \int \frac{1}{\cos^2 x}\,dx - \frac{3}{2}\int dx + \frac{1}{2}\int \cos 2x\,dx$

$= \frac{x^2}{4} - \frac{1}{2}\cdot\frac{1}{2}\int x\,d\sin 2x + \operatorname{tg} x - \frac{3}{2}x + \frac{1}{2}\cdot\frac{1}{2}\sin 2x + C =$

$= \frac{x^2}{4} - \frac{1}{4}\left[x\cdot\sin 2x - \int \sin 2x\,dx\right] + \operatorname{tg} x - \frac{3}{2}x + \frac{1}{4}\sin 2x + C =$

$= \frac{x^2}{4} - \frac{1}{4}\left[x\cdot\sin 2x + \frac{1}{2}\cos 2x\right] + \operatorname{tg} x - \frac{3}{2}x + \frac{1}{4}\sin 2x + C =$

$= \frac{x^2}{4} - \frac{1}{4}x\cdot\sin 2x - \frac{3}{2}x - \frac{1}{8}\cos 2x + \operatorname{tg} x + \frac{1}{4}\sin 2x + C$ □

Друг начин:

$\int \operatorname{tg}^2 x\cdot\sin^2 x\,dx = \int \frac{\sin^2 x}{\cos^2 x}\cdot\sin^2 x\,dx =$

$= \int \frac{\sin^4 x}{\cos^2 x}\,dx = \int \frac{(1-\cos^2 x)^2}{\cos^2 x}\,dx =$

$= \int \frac{1 - 2\cos^2 x + \cos^4 x}{\cos^2 x}\,dx =$

$= \int \frac{1}{\cos^2 x}\,dx - 2\int dx + \int \cos^2 x\,dx =$

$= \operatorname{tg} x - 2x + \int \frac{1+\cos 2x}{2}\,dx + C =$

$= \operatorname{tg} x - 2x + \frac{1}{2}\int dx + \frac{1}{2}\cdot\frac{1}{2}\int \cos 2x\,dx + C =$

$= \operatorname{tg} x - 2x + \frac{1}{2}x + \frac{1}{4}\sin 2x + C$ □